HR Attrition Overview

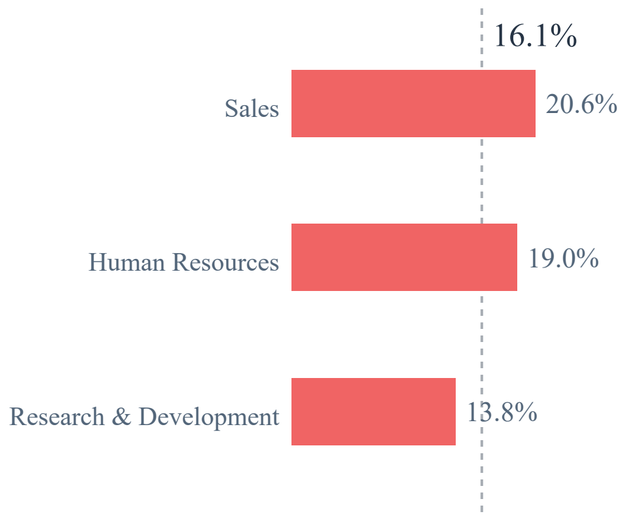
IBM HR Analytics dataset, 1470 employees

OverTime

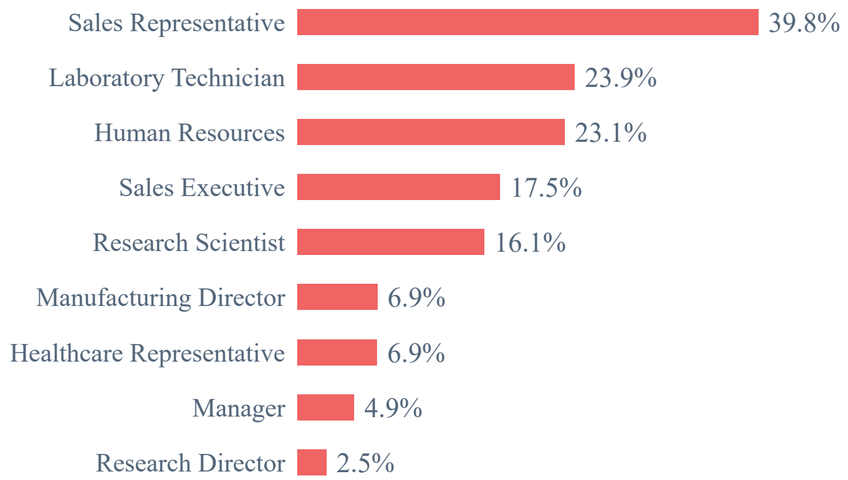
No | Yes

Total Employees	Employees Left	Attrition Rate	Avg Age	Avg Years At Company	Avg Monthly Income
1470	237	16.1%	36.9	7.0	$6,503

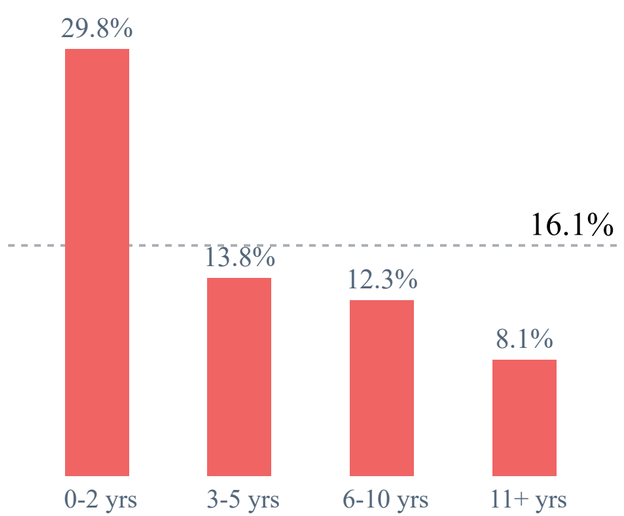
Attrition Rate
by Department



Attrition Rate
by JobRole



Attrition Rate
by TenureBand



Key insight

“Attrition is highest in the first 2 years (29.8%) and among Sales Representatives (39.8%), versus a company average of 16.1%.”

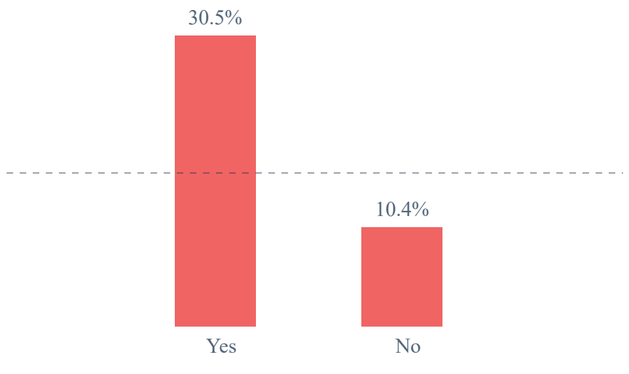
What Drives Attrition?

Each chart is paired with a statistical test

How to read this:
Dashed line = company average (16.1%). $p < 0.05$ means the pattern is unlikely to be chance. V and r show effect size: about 0.1 is small, 0.3 is moderate.

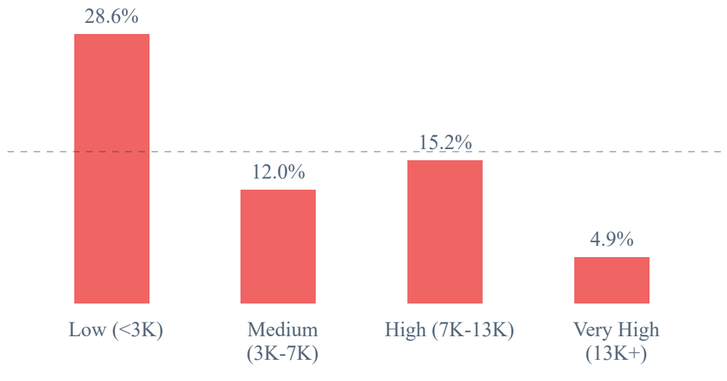
Attrition Rate by Overtime

$\chi^2(1) = 87.6 \cdot p < 0.001 \cdot V = 0.24 \cdot 3\times$ higher attrition with overtime



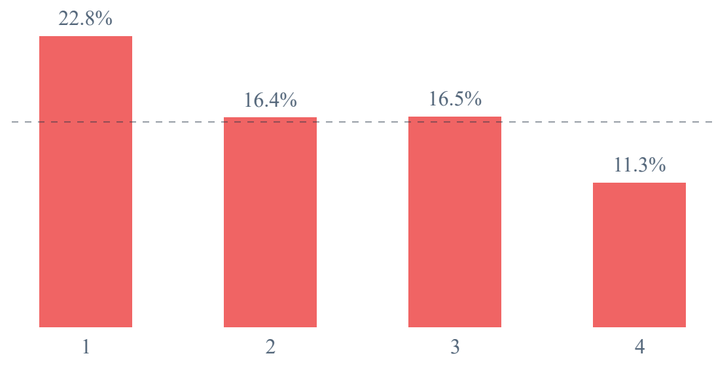
Attrition Rate by Salary Band

Mann-Whitney $p < 0.001$ · median income: leavers $3,202 vs stayers $5,204 · $r = 0.31$



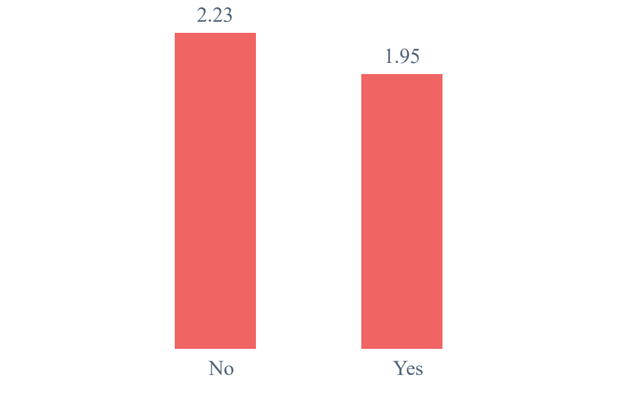
Attrition Rate by Job Satisfaction (1 = Low, 4 = High)

$\chi^2(3) = 17.5 \cdot p < 0.001 \cdot V = 0.11$ · significant but small effect



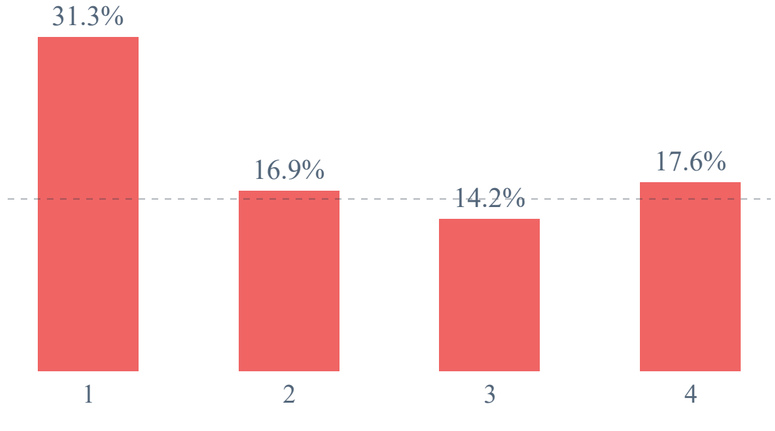
Avg Years Since Last Promotion

(No = stayed, Yes = left) Median is 1 yr for both groups · $p = 0.04 \cdot r = 0.08$ · negligible effect



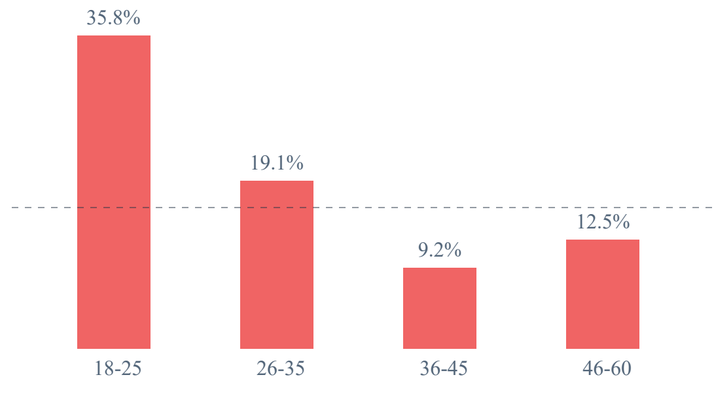
Attrition Rate by Work-Life Balance (1 = Bad, 4 = Best)

$\chi^2(3) = 16.3 \cdot p = 0.001 \cdot V = 0.11$ · level 1 has only 80 employees



Attrition Rate by Age Band

Mann-Whitney $p < 0.001$ · median age: leavers 32 vs stayers 36 · $r = 0.27$



Who is Most Likely to Leave?

Combining risk factors to find the highest-risk employee groups

How to read this:
Segments overlap. Lift = segment rate ÷ company average (16.1%). Descriptive patterns, not a prediction model.

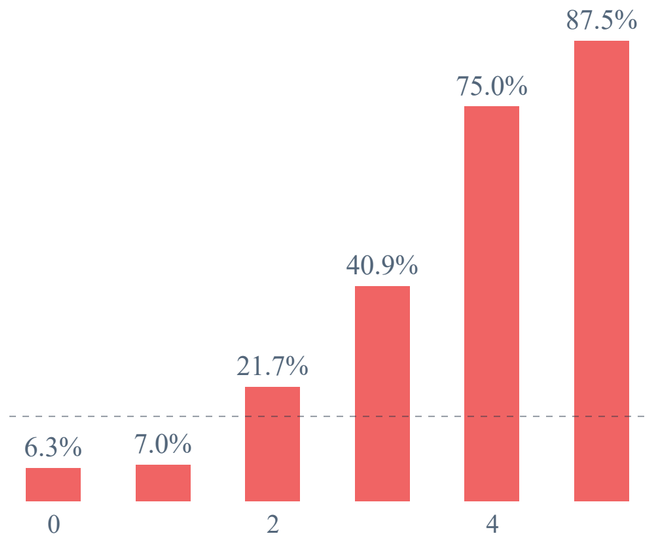
High-Risk Employee Segments

Segments overlap, and segments under 50 employees are less reliable.

Segment	Employees	Left	Attrition Rate	Lift	Share of Leavers
Overtime + Single + Income<$3K	44	31	70.5%	4.4	13.1%
Sales Reps + Overtime	24	16	66.7%	4.1	6.8%
Overtime + Income<$3K	114	64	56.1%	3.5	27.0%
Overtime + First 2 Years	104	53	51.0%	3.2	22.4%
Overtime + Single	131	65	49.6%	3.1	27.4%
Single + First 2 Years	131	57	43.5%	2.7	24.1%
Sales Representatives	83	33	39.8%	2.5	13.9%

Attrition by Number of Risk Factors

Risk factors: overtime, single, income < $3K, first 2 years, frequent travel



Independent Drivers of Attrition (Logistic Regression)

Odds ratio > 1 raises the odds of leaving, < 1 lowers them. n = 1,470, pseudo-R^2 = 0.21, AUC = 0.80.

Factor	Odds Ratio	CI95	PValue
Overtime (Yes vs No)	4.98	3.60 – 6.89	< 0.001
Single (vs married/divorced)	2.71	1.97 – 3.73	< 0.001
Frequent business travel	2.22	1.55 – 3.19	< 0.001
Distance from home (per +1 unit)	1.03	1.01 – 1.05	0.001
Age (per +1 year)	0.97	0.95 – 0.99	0.003
Monthly income (per +$1)	0.97	0.87 – 1.10	0.657
Years at company (per +1 year)	0.96	0.93 – 1.00	0.056
Job level (per +1 level)	0.77	0.48 – 1.25	0.292
Work-life balance (per +1 level)	0.76	0.61 – 0.94	0.013
Environment satisfaction (per +1 level)	0.70	0.61 – 0.81	< 0.001
Job satisfaction (per +1 level)	0.70	0.61 – 0.81	< 0.001

Key insight

- Risk stacks: attrition rises from 6.7% (0–1 risk factors) to 48.6% (3 or more).
- Overtime + single + income under $3K: 70.5% attrition, 4.4× the company average.
- Overtime is the strongest independent factor (odds ratio ≈ 5), and it is something HR can act on.

Workforce Profile

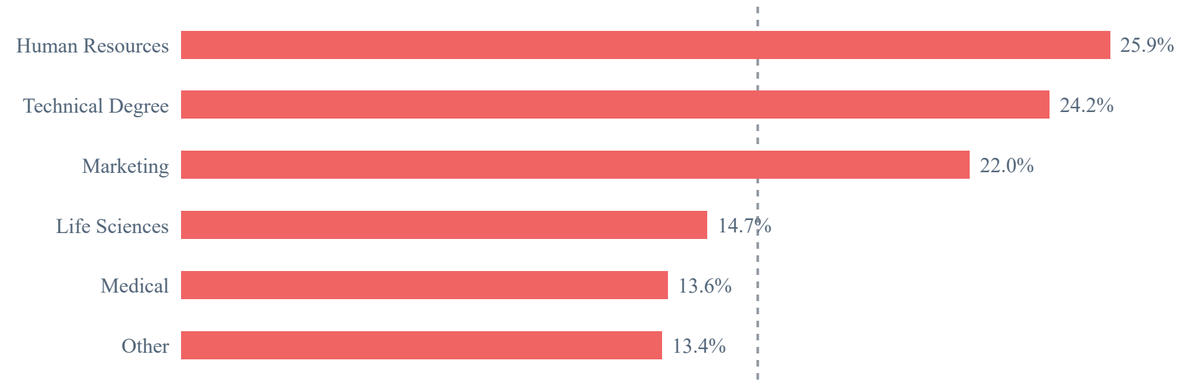
Company-wide: three junior roles (Sales Representative, Laboratory Technician, HR) are 27% of staff but 45% of leavers

Filter by department: All

Department	Employees	Left	Attrition	Avg Income	Avg Tenure (yrs)
Sales	**446**	**92**	**20.6%**	**$6,959**	**7.3**
Sales Representative	83	33	39.8%	$2,626	2.9
Sales Executive	326	57	17.5%	$6,924	7.5
Manager	37	2	5.4%	$16,987	15.2
Human Resources	**63**	**12**	**19.0%**	**$6,655**	**7.2**
Human Resources	52	12	23.1%	$4,236	5.3
Manager	11	0	0.0%	$18,089	16.3
Total	**1470**	**237**	**16.1%**	**$6,503**	**7.0**

Attrition by Education Field

$\chi^2(5) = 16.0 \cdot p = 0.007 \cdot V = 0.10 \cdot$ small effect. HR field has only 27 employees



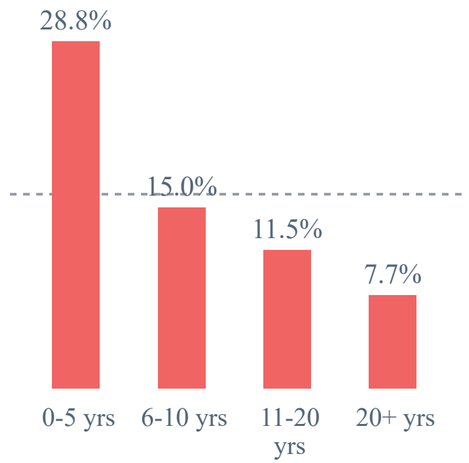
Attrition: Salary Band × Tenure Band

Cells with fewer than 30 employees are unreliable.

SalaryBand	0-2 yrs	3-5 yrs	6-10 yrs	11+ yrs
Low (<3K)	38.1% (181)	20.0% (135)	23.2% (69)	10.0% (10)
Medium (3K-7K)	23.1% (108)	12.1% (223)	9.4% (235)	4.1% (74)
High (7K-13K)	24.1% (29)	11.6% (43)	14.2% (113)	15.4% (65)
Very High (13K+)	4.2% (24)	3.0% (33)	3.2% (31)	6.2% (97)

Attrition by Total Work Experience

Mann-Whitney $p < 0.001$ · median experience: leavers 7 yrs vs stayers 10 yrs · $r = 0.31$



Attrition by Marital Status

$\chi^2(2) = 46.2 \cdot p < 0.001 \cdot V = 0.18$. Gender: 14.8% women vs 17.0% men, not significant ($p = 0.29$)

